

- less discriminative in dense representations
 - many vectors may have similar angles

Manifold

- space that is locally Euclidean → approximated by a tangent plane
 - tangent vectors: derivatives of curves
 - like a linear subspace in a sufficiently small neighborhood
- real-world data → on a much lower-dimensional manifold within a high-dimensional space
 - explaining why dimensionality reduction & representation learning works

Kernel

- function that computes inner products in an implicit feature space
 - without explicitly mapping data into high-dimensional feature space
 - implications
 - many problems become linearly separable in a higher-dimensional space
 - allow measuring similarity in a high-dimensional space efficiently
 - without paying the computational cost of explicit feature expansion
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Bayes Theorem

$$p(\theta \mid x) = \frac{p(x \mid \theta) p(\theta)}{p(x)}$$

- reveal how to update our belief about model parameters θ after observing data x
 - $p(x \mid \theta)$: likelihood
 - $p(\theta)$: prior

- $p(\theta \mid x)$: posterior
- $p(x)$: marginal likelihood (evidence)

Likelihood

- function of parameters
 - measure how well a set of parameters θ explains the observed data
 - not a probability distribution over the data x
- Gaussian model: $x \sim \mathcal{N}(\mu, \sigma^2)$

$$\text{Likelihood: } p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Maximum Likelihood Estimation (MLE)

- choose parameters that maximizes the likelihood of the observed data
 - without any prior belief
 - how to define the objective
 - not optimization method itself (e.g., gradient descent)
- usually maximize log-likelihood: $\arg \max_{\theta} \log p(x \mid \theta)$
 - usually corresponds to minimizing a negative log-likelihood loss
 - e.g., MSE, cross-entropy

Maximum A Posterior (MAP)

- similar to MLE but incorporate a prior belief about parameters θ
 - also likely under a prior distribution (regularization)
 - prior: uniform $\rightarrow$ MAP: MLE
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- concentrating on a few dominant directions
 - keeping top singular components ($\approx$ SVD perspective)
 - capturing the most important adaptation directions efficiently
 - memory efficient training
 - fine-tuning large pretrained models
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Probability Distribution

- modeling uncertainty of random variables
 - mapping outcomes to probabilities
- discrete v.s. continuous distributions
 - probability mass function (PMF) v.s. probability density function (PDF)

Bernoulli Distribution

$$P(X = x) = p^x(1 - p)^{1-x}, \quad x \in \{0, 1\}$$

- single binary trial (e.g., success / failure)
 - $X \in \{0, 1\}$
 - success probability p

Binomial Distribution

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

- number of successes k in a fixed number of trials n
 - independent Bernoulli trials
 - success probability p
- statistics

- mean: np
- variance: $np(1 - p)$
- normal approximation when n large

Multinomial Distribution

$$P(\mathbf{X} = \mathbf{x}) = \frac{n!}{x_1! \cdots x_K!} \prod_{i=1}^K p_i^{x_i}$$

- generalization of Bernoulli / Binomial to multiple categories
- used in multi-class classification
 - categorical likelihood
 - softmax output

Normal(Gaussian) Distribution

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- mean μ
- variance σ^2
- commonly used as a noise model in regression

Exponential Distribution

단위 시간 당 λ 번의 사건 발생

$$p(x) = \lambda e^{-\lambda x}, \quad x \geq 0$$

다음 사건이 언제 발생할지 (걸리는 시간)

- models waiting time until the next event
 - time between events in a Poisson process
 - rate $\lambda > 0$
- mean: $1/\lambda$
- variance: $1/\lambda^2$
- memoryless property
 - $P(X > s + t \mid X > s) = P(X > t)$

- future waiting time independent of the past

Poisson Distribution

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

단위 시간에 k 번 사건이 발생할 확률

- event counts over time or space
 - unknown number of trials
- $X \in \{0, 1, 2, \dots\}$
- event rate λ 단위 시간에 평균으로 λ 번 사건 발생
- Binomial limit when $n \rightarrow \infty, p \rightarrow 0$
- independent event assumption
- mean = variance = λ

Gamma Distribution

$$p(x) = \frac{\lambda^k}{\Gamma(k)} x^{k-1} e^{-\lambda x}, \quad x \geq 0$$

- generalization of the Exponential distribution
- models waiting time until the k -th event
 - shape k
 - rate λ
- mean: k/λ
- variance: k/λ^2
- when $k = 1 \rightarrow$ Gamma = Exponential

Central Limit Theorem

$X_1, \dots, X_n$ i.i.d.

$$\mathbb{E}[X_i] = \mu, \text{Var}(X_i) = \sigma^2 < \infty$$

- number of samples $n \uparrow \rightarrow$ distribution of the sample mean after normalization converges to a Gaussian distribution
 - ■ $\bar{X} \approx \mathcal{N}(\mu, \sigma^2/n)$ for large n
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Linear Regression

- models the relationship between input features and a target variable
 - assumes a linear relationship between inputs and output
- model
 - $y = Xw + \epsilon$
 - X : input features
 - w : weight vector
 - ϵ : noise term (usually assumed Gaussian)
- objective
 - $\mathcal{L}(w) = \frac{1}{N} \sum_{i=1}^N (y_i - x_i^\top w)^2$
 - minimize prediction error between model output and ground-truth targets
 - typically using mean squared error (MSE)
- interpretation
 - each weight w_j represents the contribution of feature j to the output
 - learns a hyperplane in feature space
 - gradient-based optimization
 - compute gradient of MSE w.r.t. w
 - $\nabla_w \mathcal{L} = -\frac{2}{N} X^\top (y - Xw)$
 - update using gradient descent
 - $w_{t+1} = w_t - \eta \nabla_w \mathcal{L}(w_t)$